



TerpSoccer

BACKGROUND

Who are the end users?

- Coaches and training staff
- Players
- Analysts

Data Source

- UMD Soccer Website - <https://umterps.com/sports/mens-soccer/stats>

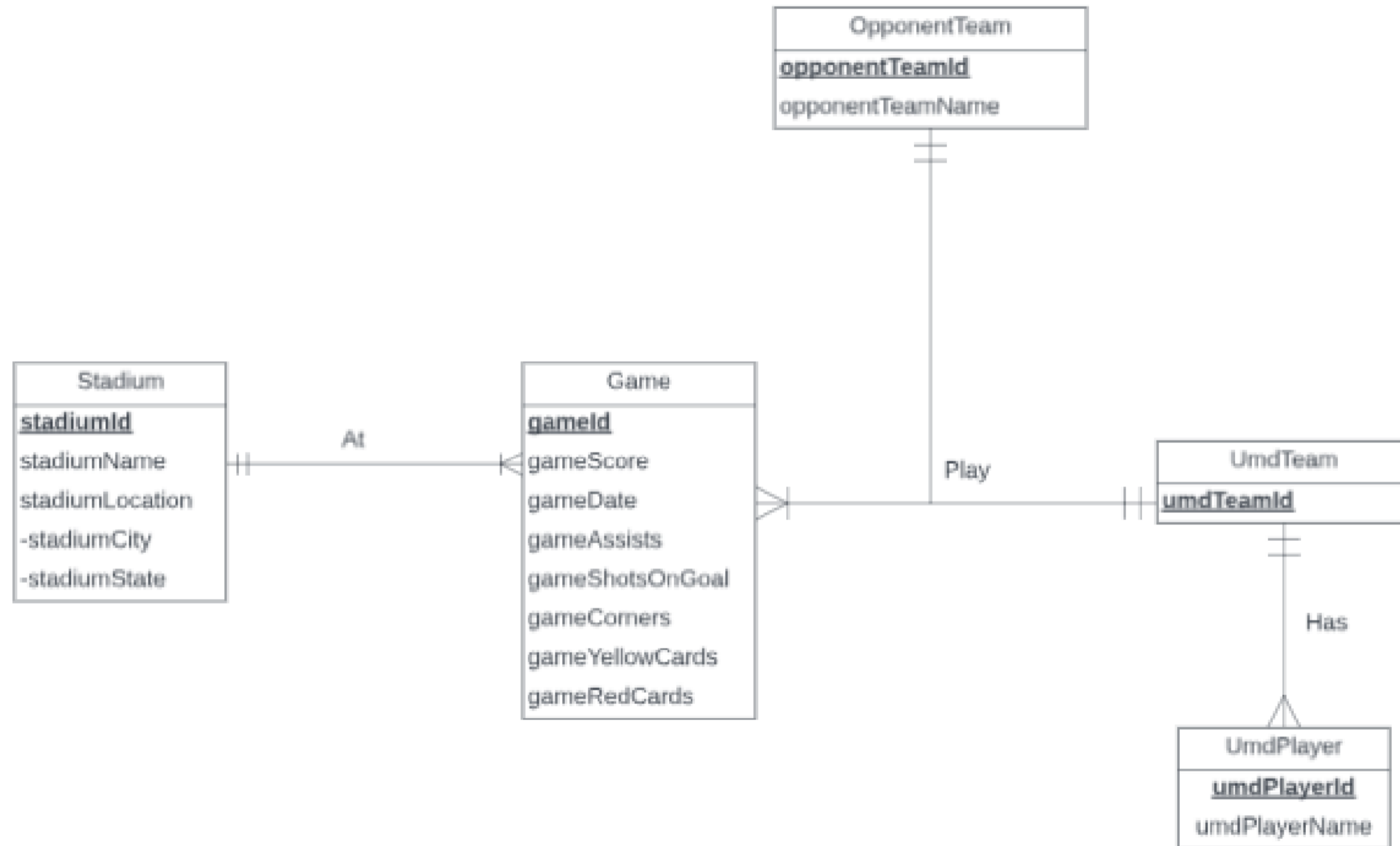
Mission Statements and Objectives

Our mission is to structure the data available on UMD soccer stats, and provide insights on efficient team management, game results and game statistics to enable data driven decision making and performance improvements.

Mission Objectives:

- Track game statistics and the distribution of goals in first and second half of each game.
- Record opponent details as well as location details of each game
- Maintain data on UMD Team's players and coaches which will provide key insights into identification of key contributors and the effectiveness of the management.
- Analyse and query the database to identify performance trends, enabling data-driven strategies to enhance the team's future performance.

CONCEPTUAL DATABASE DESIGN - ER DIAGRAM



LOGICAL DATABASE DESIGN - RELATIONAL SCHEMA

OpponentTeam (**opponentTeamId**, opponentTeamName)

Stadium (**stadiumId**, stadiumName, stadiumCity, stadiumState)

UmdTeam (**umdTeamId**)

UmdPlayer (**umdPlayerId**, umdPlayerName, *umdTeamId*)

Game (**gameId**, gameScore, gameDate, gameAssists, gameShotsOnGoal, gameCorners, gameYellowCards, gameRedCards, *stadiumId*, *opponentTeamId*, *umdTeamId*)

Play (**gameId**, *opponentTeamId*, *umdTeamId*)

PHYSICAL DATABASE DESIGN

```
CREATE TABLE OpponentTeam (  
  opponentTeamId CHAR(5) NOT NULL,  
  opponentTeamName VARCHAR(20) NOT NULL,  
  CONSTRAINT pk_Team_teamId PRIMARY KEY (opponentTeamId),  
)
```

```
CREATE TABLE Stadium (  
  stadiumId CHAR(5) NOT NULL,  
  stadiumName VARCHAR(30),  
  stadiumCity VARCHAR(20) NOT NULL,  
  stadiumState VARCHAR(20) NOT NULL,  
  CONSTRAINT pk_Stadium_stadiumId PRIMARY KEY (stadiumId)  
)
```

PHYSICAL DATABASE DESIGN

```
CREATE TABLE UmdTeam (  
  umdTeamId CHAR(4) NOT NULL,  
  CONSTRAINT pk_UmdTeam_umdTeamId PRIMARY KEY (umdTeamId)  
)
```

```
CREATE TABLE UmdPlayer (  
  umdPlayerId CHAR(6) NOT NULL,  
  umdPlayerName VARCHAR(50),  
  umdTeamId CHAR(4) NOT NULL,  
  CONSTRAINT pk_UmdPlayer_playerId PRIMARY KEY (umdPlayerId),  
  CONSTRAINT fk_UmdPlayer_umdTeam FOREIGN KEY (umdTeamId)  
    REFERENCES UmdTeam (umdTeamId)  
  ON DELETE CASCADE ON UPDATE CASCADE  
)
```

PHYSICAL DATABASE DESIGN

```
CREATE TABLE Game(  
  gameId CHAR(6) NOT NULL,  
  gameScore VARCHAR(10) NOT NULL,  
  opponentTeamName VARCHAR(20) NOT NULL,  
  gameDate DATE,  
  gameAssists VARCHAR(5),  
  gameShotsOnGoal VARCHAR(5),  
  gameCorners VARCHAR(5),  
  gameYellowCards VARCHAR(5),  
  gameRedCards VARCHAR(5),  
  stadiumId CHAR(5) NOT NULL,  
  umdTeamId CHAR(4) NOT NULL,  
  opponentTeamId CHAR(5) NOT NULL,  
  CONSTRAINT pk_Game_gameId PRIMARY KEY (gameId),  
  CONSTRAINT fk_Game_stadiumId FOREIGN KEY (stadiumId)  
    REFERENCES Stadium (stadiumId)  
    ON DELETE CASCADE ON UPDATE CASCADE,  
  CONSTRAINT fk_Game_umdTeamId FOREIGN KEY (umdTeamId)  
    REFERENCES UmdTeam (umdTeamId)  
    ON DELETE CASCADE ON UPDATE CASCADE,  
  CONSTRAINT fk_Game_opponentTeamId FOREIGN KEY (opponentTeamId)  
    REFERENCES OpponentTeam (opponentTeamId)  
    ON DELETE CASCADE ON UPDATE CASCADE  
)
```

```
CREATE TABLE Play(  
  gameId CHAR(6) NOT NULL,  
  umdTeamId CHAR(4) NOT NULL,  
  opponentTeamId CHAR(5) NOT NULL,  
  CONSTRAINT pk_Play_gameId PRIMARY KEY (gameId),  
  CONSTRAINT fk_Play_gameId FOREIGN KEY (gameId)  
    REFERENCES Game (gameId)  
    ON DELETE NO ACTION ON UPDATE NO ACTION,  
  CONSTRAINT fk_Play_umdTeamId FOREIGN KEY (umdTeamId)  
    REFERENCES umdTeam (umdTeamId)  
    ON DELETE NO ACTION ON UPDATE NO ACTION,  
  CONSTRAINT fk_Play_opponentTeamId FOREIGN KEY (opponentTeamId)  
    REFERENCES OpponentTeam (opponentTeamId)  
    ON DELETE NO ACTION ON UPDATE NO ACTION  
)
```


CREATING A VIEW

Game results view with stadium name

```
-- Game results view with stadium name and game result
DROP VIEW IF EXISTS GameResultsView;
GO
CREATE VIEW GameResultsView AS
SELECT
    g.gameScore AS 'Game Score',
    CASE
        WHEN CAST(SUBSTRING(g.gameScore, 1, CHARINDEX('-', g.gameScore) - 1) AS INT) >
            CAST(SUBSTRING(g.gameScore, CHARINDEX('-', g.gameScore) + 1, LEN(g.gameScore)) AS INT)
        THEN 'Win'
        WHEN CAST(SUBSTRING(g.gameScore, 1, CHARINDEX('-', g.gameScore) - 1) AS INT) <
            CAST(SUBSTRING(g.gameScore, CHARINDEX('-', g.gameScore) + 1, LEN(g.gameScore)) AS INT)
        THEN 'Loss'
        ELSE 'Tie'
    END AS 'Game Result',
    CASE
        WHEN g.stadiumId = '10001'
        THEN 'Home'
        ELSE 'Away'
    END AS 'Location',
    g.opponentTeamName AS 'Opponent',
    g.gameDate AS 'Game Date',
    g.gameAssists AS 'Assists',
    g.gameShotsOnGoal AS 'Shots On Goal',
    g.gameCorners AS 'Corners',
    g.gameYellowCards AS 'Yellow Cards',
    g.gameRedCards AS 'Red Cards',
    s.stadiumName AS 'Stadium Name'
FROM Game g JOIN Stadium s
ON g.stadiumId = s.stadiumId
```

BUSINESS TRANSACTIONS

```
-- Which games had the poorest discipline i.e. many yellow or red cards?  
SELECT *  
FROM GameResultsView  
WHERE [Yellow Cards] <> '0/0' OR [Red Cards] <> '0/0'  
ORDER BY [Game Date];
```

	Game Score	Game Result	Location	Opponent	Game Date	Assists	Shots On Goal	Comers	Yellow Cards	Red Cards	Stadium Name
1	0-0	Tie	Away	Notre Dame	2015-08-28	0/0	4/4	8/4	1/1	0/0	Armstrong Stadium
2	5-0	Win	Home	HARTFORD	2016-08-29	6/0	10/0	12/1	0/2	0/0	Ludwig Field
3	2-1	Win	Home	Georgetown	2016-09-02	1/2	7/4	5/3	0/1	0/0	Ludwig Field
4	4-1	Win	Home	USF	2016-09-05	6/0	6/3	3/1	1/4	0/1	Ludwig Field
5	1-1	Tie	Away	Indiana	2016-09-09	1/1	4/5	5/9	1/1	0/0	Armstrong Stadium
6	2-1	Win	Away	Rutgers	2016-09-13	1/0	12/1	7/2	1/1	0/0	Yurcak Field
7	4-0	Win	Away	Northwestern	2016-09-30	8/0	9/0	5/4	0/2	0/0	Martin Stadium
8	2-0	Win	Home	Ohio State	2016-10-07	3/0	8/2	3/3	1/2	0/0	Ludwig Field
9	1-0	Win	Away	West Virginia	2016-10-10	1/0	2/8	7/1	2/1	0/0	Dick Dlesk Stadium
10	3-2	Win	Home	Wisconsin	2016-10-14	4/3	11/7	7/5	1/3	0/0	Ludwig Field

BUSINESS TRANSACTIONS

```
--What are the total number of games played in each year?  
SELECT ut.umdTeamId AS 'Game Year', COUNT(pl.gameId) AS 'Number of Games Played'  
FROM Play pl  
JOIN UmdTeam ut ON pl.umdTeamId = ut.umdTeamId  
GROUP BY ut.umdTeamId;
```

	Game Year	Number of Games Played
1	2015	1
2	2016	21
3	2017	19
4	2018	16
5	2019	21
6	2020	11
7	2021	18
8	2022	20
9	2023	15
10	2024	13

BUSINESS TRANSACTIONS

```
--What are the total number of wins and number of losses per year for the UMD team?
SELECT
    YEAR(g.gameDate) AS 'Game Year',
    SUM(CASE
        WHEN CAST(LEFT(g.gameScore, CHARINDEX('-', g.gameScore) - 1) AS INT) >
            CAST(RIGHT(g.gameScore, LEN(g.gameScore) - CHARINDEX('-', g.gameScore)) AS INT)
        THEN 1 ELSE 0
    END) AS 'Total Wins',
    SUM(CASE
        WHEN CAST(LEFT(g.gameScore, CHARINDEX('-', g.gameScore) - 1) AS INT) <
            CAST(RIGHT(g.gameScore, LEN(g.gameScore) - CHARINDEX('-', g.gameScore)) AS INT)
        THEN 1 ELSE 0
    END) AS 'Total Loss',
    SUM(CASE
        WHEN CAST(LEFT(g.gameScore, CHARINDEX('-', g.gameScore) - 1) AS INT) =
            CAST(RIGHT(g.gameScore, LEN(g.gameScore) - CHARINDEX('-', g.gameScore)) AS INT)
        THEN 1 ELSE 0
    END) AS 'Total Ties'
FROM Game g
GROUP BY YEAR(g.gameDate)
ORDER BY [Game Year];
```

	Game Year	Total Wins	Total Loss	Total Ties
1	2015	0	0	1
2	2016	18	1	2
3	2017	10	5	4
4	2018	7	6	3
5	2019	11	8	2
6	2021	19	7	3
7	2022	11	4	5
8	2023	4	8	3
9	2024	8	1	4

BUSINESS TRANSACTIONS

```
-- How many games were won,loss and tied at different stadiums?
SELECT
  s.stadiumId,
  s.stadiumName,
  SUM(CASE
    WHEN CAST(LEFT(g.gameScore, CHARINDEX('-', g.gameScore) - 1) AS INT) >
      CAST(RIGHT(g.gameScore, LEN(g.gameScore) - CHARINDEX('-', g.gameScore)) AS INT) THEN 1
    ELSE 0
  END) AS 'Total Wins',
  SUM(CASE
    WHEN CAST(LEFT(g.gameScore, CHARINDEX('-', g.gameScore) - 1) AS INT) <
      CAST(RIGHT(g.gameScore, LEN(g.gameScore) - CHARINDEX('-', g.gameScore)) AS INT) THEN 1
    ELSE 0
  END) AS 'Total loss',
  SUM(CASE
    WHEN CAST(LEFT(g.gameScore, CHARINDEX('-', g.gameScore) - 1) AS INT) =
      CAST(RIGHT(g.gameScore, LEN(g.gameScore) - CHARINDEX('-', g.gameScore)) AS INT) THEN 1
    ELSE 0
  END) AS 'Total Ties'
FROM Game g
JOIN Stadium s ON g.stadiumId = s.stadiumId
GROUP BY s.stadiumId, stadiumName
```

	stadiumId	stadiumName	Total Wins	Total loss	Total Ties
1	10001	Ludwig Field	61	22	15
2	10002	Audi Field	0	1	0
3	10003	U-M Socce...	2	1	0
4	10004	Miller Family...	1	0	0
5	10005	Lanny and ...	2	0	1
6	10006	Armstrong ...	1	2	3
7	10007	Berman Field	0	1	0
8	10008	Shaw Field	1	1	1
9	10009	Jesse Owe...	3	1	1
10	10010	Dan McCli...	1	2	0
11	10011	Grant Stadi...	1	0	0
12	10012	Old College...	1	0	1
13	10013	Jeffrey Field	1	3	0
14	10014	Kl'ckner St...	1	1	0
15	10015	Holuba Hall	0	1	0
16	10016	Yurcak Field	3	0	0
17	10017	Legacy Ce...	0	1	0
18	10018	Walis Anne...	0	1	0
19	10019	DeMartin St...	0	0	2
20	10020	Spry Stadium	0	1	0

BUSINESS TRANSACTIONS

```
-- Which opponent teams have frequently beaten UMD Team?
SELECT g.opponentTeamName AS 'Opponent Team Name', COUNT(*) AS 'Number of Losses'
FROM Game g
WHERE
    CAST(SUBSTRING(g.gameScore, 1, CHARINDEX('-', g.gameScore) - 1) AS INT) <
    CAST(SUBSTRING(g.gameScore, CHARINDEX('-', g.gameScore) + 1, LEN(g.gameScore)) AS INT)
GROUP BY g.opponentTeamName
ORDER BY 'Number of Losses' DESC;
```

	Opponent Team Name	Number of Losses
1	Indiana	5
2	Michigan	4
3	Georgetown	4
4	Wisconsin	4
5	Virginia	2
6	Penn	2
7	Penn St.	2
8	Northwestern	2
9	UCLA	2
10	VCU	1

BUSINESS TRANSACTIONS

```
-- How was the performance of UMDTeam against opponents with consistently strong defensive records?
SELECT g.opponentTeamName AS 'Opponent Team Name', COUNT(*) AS 'Number of games UMDTeam Failed To Score'
FROM Game g
WHERE
    CAST(SUBSTRING(g.gameScore, 1, CHARINDEX('-', g.gameScore) - 1) AS INT) = 0
GROUP BY g.opponentTeamName
ORDER BY 'Number of games UMDTeam Failed To Score' DESC;
```

	Opponent Team Name	Number of games UMDTeam Failed To Score
1	Georgetown	5
2	Indiana	3
3	Wisconsin	3
4	Michigan State	2
5	Rutgers	2
6	Northwestern	2
7	Virginia	2
8	Wake Forest	2
9	Washington	1
10	Notre Dame	1

BUSINESS TRANSACTIONS

```
--Which were the games where the UMDTeam had high offensive output but still lost?
SELECT g.gameId AS 'Game Id', g.opponentTeamName AS 'Opponent Team Name', g.gameScore AS 'Game Score',
       g.gameShotsOnGoal AS 'Shots on Goal', g.gameAssists AS 'Game Assists'
FROM Game g
WHERE
    ABS(TRY_CAST(SUBSTRING(g.gameScore, 1, CHARINDEX('-', g.gameScore) - 1) AS INT)
        - TRY_CAST(SUBSTRING(g.gameScore, CHARINDEX('-', g.gameScore) + 1, LEN(g.gameScore)) AS INT)) = 1
AND TRY_CAST(SUBSTRING(g.gameScore, 1, CHARINDEX('-', g.gameScore) - 1) AS INT) <
    TRY_CAST(SUBSTRING(g.gameScore, CHARINDEX('-', g.gameScore) + 1, LEN(g.gameScore)) AS INT)
ORDER BY g.gameId DESC;
```

	Game Id	Opponent Team Name	Game Score	Shots on Goal	Game Assists
1	202313	Indiana	1-2	5/3	2/3
2	202309	Rutgers	1-2	11/3	2/1
3	202307	Wisconsin	0-1	1/2	0/0
4	202306	Penn	0-1	5/5	0/0
5	202304	Virginia	1-2	6/7	1/0
6	202301	Missouri	0-1	1/5	0/1
7	202220	Cornell	1-2	4/6	1/2
8	202218	Indiana	1-2	2/4	1/2
9	202213	High Point	1-2	5/2	0/1
10	202118	LIU	0-1	5/2	0/0

BUSINESS TRANSACTIONS

```
-- Which were the matches that were extremely close in terms of score? (Focus on games where the difference in
-- score was only 1 goal, indicating close games)
SELECT g.gameId AS 'Game Id', g.opponentTeamName AS 'Opponent Team Name', g.gameDate AS 'Game Date', g.gameScore AS 'Game Score'
FROM Game g
WHERE ABS(TRY_CAST(LEFT(g.gameScore, CHARINDEX('-', g.gameScore) - 1) AS INT) -
          TRY_CAST(RIGHT(g.gameScore, LEN(g.gameScore) - CHARINDEX('-', g.gameScore)) AS INT)) = 1
ORDER BY g.gameDate DESC;
```

	Game Id	Opponent Team Name	Game Date	Game Score
1	202412	Washington	2024-10-11	3-2
2	202411	Michigan St.	2024-10-05	2-1
3	202409	Northwestern	2024-09-27	2-1
4	202408	Indiana	2024-09-20	1-0
5	202405	Brown	2024-09-06	2-1
6	202313	Indiana	2023-10-20	1-2
7	202310	Bowling Green	2023-10-07	1-0
8	202309	Rutgers	2023-10-03	1-2
9	202307	Wisconsin	2023-09-24	0-1
10	202306	Penn	2023-09-19	0-1

THANK YOU